HARMONIC NODAL GEOMETRY, ZERO-CURVE CURVATURE, AND THE TRANSVERSE INTEGRAL IDENTITY FOR RIEMANN'S Ξ-FUNCTION

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ABSTRACT. For any entire function $f(t-\mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$ real on $\mathbb{R}$, Cauchy–Riemann orthogonality gives $\frac{\mathrm{d}|R|}{\mathrm{d}s} = \pm|\nabla I| = \pm|f'(t-\mathrm{i}d)|$ along smooth imaginary nodal curves $\{I(d,t) = 0\}$ away from critical points $f'(z) = 0$, while $|f(t-\mathrm{i}d)|^2 = |f(t)|^2 + 2\int_0^d u\,\widetilde{W}_f(u,t)\,\mathrm{d}u$ holds identically. Specializing to Riemann's $\Xi(t-\mathrm{i}d) = \xi(\frac{1}{2}+d+\mathrm{i}t)$, we clarify that global nodal monotonicity requires both critical-line Laguerre positivity $L_1(\Xi;\tau_k) > 0$ and the critical-point-free hypothesis $\Xi'(z) \neq 0$ in $d > 0$ (equivalent to the Riemann Hypothesis by Lagarias, 1999). Each real zero curve $R(d,t) = 0$ departs orthogonally from a simple zero γ_k with vertex curvature $\kappa(\gamma_k) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$, matching the de Bruijn–Newman zero-velocity ratio of Csordas, Smith, and Varga (1994). Under the canonical normalization $W(d,t) = 16\int_0^\infty y\sinh(2dy)\,I_{\mathrm{Wig}}(y,t)\,\mathrm{d}y$, transverse integration maps $\mathrm{sinc}_{\mathrm{hyp}}(2uy)$ to $\mathrm{sinc}^2_{\mathrm{hyp}}(dy)$ with filter ratio $\frac{\tanh(dy)}{dy} \in (0,1)$, verified alongside explicit counterexample mechanisms across $\gamma_1,\dots,\gamma_{15}$ and $\Omega_1,\dots,\Omega_{25}$.

1. INTRODUCTION

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ denote Riemann's completed zeta function, and write $s = \frac{1}{2}+d+\mathrm{i}t$ with transverse offset $d = \sigma - \frac{1}{2} \in \mathbb{R}$ and spectral height $t \in \mathbb{R}$. Setting $z = -\mathrm{i}(s-\frac{1}{2}) = t - \mathrm{i}d$, Riemann's entire function $\Xi(z) := \xi(\frac{1}{2}+\mathrm{i}z) = R(d,t) + \mathrm{i}I(d,t)$ admits the classical Fourier–Laplace representation [18, 16, 24]

$$\begin{aligned}\Xi(t-\mathrm{i}d) &= 8\int_0^\infty \Phi(u)\cosh(2du)\cos(2tu)\,\mathrm{d}u + \mathrm{i}\,8\int_0^\infty \Phi(u)\sinh(2du)\sin(2tu)\,\mathrm{d}u \\ &= 4\int_{-\infty}^\infty \Phi(u)\,\mathrm{e}^{2(d+\mathrm{i}t)u}\,\mathrm{d}u,\end{aligned} \tag{1.1}$$

where the even, rapidly decaying Jacobi–Riemann kernel $\Phi(u) = \Phi(-u) > 0$ on $\mathbb{R}$ is given by

$$\Phi(u) := \sum_{n=1}^\infty \left(2\pi^2 n^4 \mathrm{e}^{9u} - 3\pi n^2 \mathrm{e}^{5u}\right)\exp\left(-\pi n^2 \mathrm{e}^{4u}\right). \tag{1.2}$$

In the canonical normalization fixed in Papers I and II [12, 13], the transverse Wronskian $W(d,t) := \frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2 = R\,\partial_t I - I\,\partial_t R$ and its parity-reduced quotient $\widetilde{W}(d,t) := W(d,t)/d$ take the exact Wigner phase-space forms

$$W(d,t) = 16\int_0^\infty y\sinh(2dy)\,I_{\mathrm{Wig}}(y,t)\,\mathrm{d}y, \qquad \widetilde{W}(d,t) = 32\int_0^\infty y^2\,\mathrm{sinc}_{\mathrm{hyp}}(2dy)\,I_{\mathrm{Wig}}(y,t)\,\mathrm{d}y, \tag{1.3}$$

where $\mathrm{sinc}_{\mathrm{hyp}}(w) := \frac{\sinh w}{w} \geq 1$ and the even correlation kernel $I_{\mathrm{Wig}}(y,t) = 2\pi\,\mathcal{W}_\Phi(y/2,2t)$ is defined by

$$I_{\mathrm{Wig}}(y,t) := \int_{-\infty}^\infty \Phi\left(\frac{x+y}{2}\right)\Phi\left(\frac{x-y}{2}\right)\cos(2tx)\,\mathrm{d}x. \tag{1.4}$$

The geometric study of the critical strip via real and imaginary zero loci goes back to Speiser [20], Spira [21, 22, 23], Hinkkanen [8], Lagarias [10], and the X-ray investigations of Arias de Reyna [1]. Several distinctions from the classical literature must be stated clearly at the outset:

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(1) *Universal harmonic identities versus RH-equivalent hypotheses:* Any entire function $f(t-\mathrm{id}) = R(d,t) + \mathrm{i}I(d,t)$ that is real on $\mathbb{R}$—including both $\Xi(z)$ and counterexamples without an Euler product such as the Davenport–Heilbronn function $\Xi_{\mathrm{DH}}(z)$ [5]—satisfies $\Delta R = \Delta I = 0$, the critical-line Dirichlet/Neumann conditions $I(0,t) \equiv \partial_d R(0,t) \equiv 0$, the arc-length differential identity $\frac{\mathrm{d}|R|}{\mathrm{d}s} = \pm|\nabla I| = \pm|f'(t-\mathrm{id})|$ along smooth curves $\{I(d,t)=0\}$ away from critical points $f'(z)=0$, and the transverse integral identity $|f(t-\mathrm{id})|^2 = |f(t)|^2 + 2\int_0^d u\,\widetilde{W}_f(u,t)\,\mathrm{d}u$. Assuming globally that $\Xi'(t-\mathrm{id}) \neq 0$ in $d>0$ is, by the Gauss–Lucas argument of Lagarias [10], equivalent to the Riemann Hypothesis (RH) itself and therefore cannot serve as an unconditional input.

(2) *Speiser's theorem for $\zeta'(s)$ versus critical points of $\Xi(z)$:* Speiser [20] proved that RH is equivalent to $\zeta'(s) \neq 0$ in the *left* half-strip $0 < \sigma < \frac{1}{2}$, whereas $\zeta'(s)$ possesses infinitely many zeros in the right half-plane $\frac{1}{2} < \sigma < 3$ [21, 22, 23]. For the completed function $\xi(s) = \Xi(t-\mathrm{id})$, by contrast, functional symmetry makes $\Xi'(t-\mathrm{id}) \neq 0$ in $d>0$ ($\sigma > \frac{1}{2}$) and $\mathrm{Re}(\xi'(s)/\xi(s)) > 0$ in $d>0$ equivalent to RH [8, 10].

(3) *Prior art on zero-curve curvature and finite-height zero verification:* At a simple zero γ_k of $\Xi(t)$, the ratio $\Xi''(\gamma_k)/\Xi'(\gamma_k)$ appearing below as the vertex curvature $\kappa(\gamma_k) = t''_{\mathcal{R}_k}(0)$ of the real nodal curve $R(d,t)=0$ coincides with the classical zero-velocity expression governing Lehmer-pair dynamics under the de Bruijn–Newman heat flow studied by Csordas, Smith, and Varga [3], while the boundary concavity $c_1(t) = (\Xi'(t))^2 - \Xi(t)\Xi''(t) = L_1(\Xi;t)$ is the classical first Laguerre expression [9, 4]. Likewise, simplicity and critical-line location of all zeros up to $t \le 1000$ were established long ago by Gram [7] and Titchmarsh and Comrie [25] and now extend to $t \le 3\cdot 10^{12}$ by Platt and Trudgian [15]; our numerical evaluations in Section 5 serve strictly to illustrate the exact local jets, chamber geometry, and counterexample mechanisms.

With these attributions and logical boundaries fixed, the main results of this note are summarized as follows.

Theorem 1.1 (Harmonic Nodal Geometry, Local Zero-Curve Jets, and Transverse Integral Identity)**.** *Let $f(t-\mathrm{id}) = R_f(d,t) + \mathrm{i}I_f(d,t)$ be an entire function that is real-valued on $\mathbb{R}$, with transverse Wronskian $W_f(d,t) := \frac{1}{2}\partial_d|f(t-\mathrm{id})|^2$ and parity-reduced quotient $\widetilde{W}_f(d,t) := W_f(d,t)/d$.*

(i) ***Universal harmonic structure and two-mechanism dichotomy for off-critical zeros:*** *Everywhere on $\mathbb{R}^2$, $\Delta R_f = \Delta I_f = 0$, $I_f(0,t) \equiv \partial_d R_f(0,t) \equiv 0$, and no branch of $I_f(d,t)=0$ can enclose a non-empty bounded domain (unless f is constant). Transverse imaginary nodal walls $\Gamma \subset \{I_f(d,t)=0,\, d>0\}$ depart orthogonally from $d=0$ exclusively at real critical points $f'(\tau)=0$, with vertex curvature $\kappa_\Gamma(\tau) = t''_\Gamma(0) = \frac{f'''(\tau)}{3f''(\tau)}$ whenever $f''(\tau)\neq 0$. Along any smooth segment of Γ parameterized by arc length $s>0$ departing from $(0,\tau)$ with $f(\tau)\neq 0$ and free of interior critical points $f'(z)=0$,*

$$\frac{\mathrm{d}}{\mathrm{d}s}|R_f(d(s),t(s))| = \mathrm{sgn}\left(L_1(f;\tau)\right)|\nabla I_f(d(s),t(s))| = \mathrm{sgn}\left(L_1(f;\tau)\right)|f'(t(s)-\mathrm{id}(s))|, \tag{1.5}$$

where $L_1(f;\tau) = -f(\tau)f''(\tau)$. Consequently, an off-critical zero $f(t_*-\mathrm{id}_*)=0$ ($d_*>0$) on a nodal wall connected to $(0,\tau)$ occurs only if either $L_1(f;\tau)\le 0$ on $d=0$ or Γ encounters an interior critical point $f'(z_{\mathrm{crit}})=0$ in $d>0$.

(ii) ***Local zero-curve curvature jet and Euler–Stirling right boundary in $\sigma>1$:*** *Around any simple real zero γ_k of $f(t)$ ($f(\gamma_k)=0$, $f'(\gamma_k)\neq 0$), the real nodal curve $\mathcal{R}_k := \{(d, t_{\mathcal{R}_k}(d)) : R_f(d,t)=0\}$ departs orthogonally from $d=0$ with Taylor expansion*

$$t_{\mathcal{R}_k}(d) - \gamma_k = \frac{f''(\gamma_k)}{2\,f'(\gamma_k)}\,d^2 + c_{\mathrm{quart}}(\gamma_k)\,d^4 + \mathcal{O}(d^6), \qquad \kappa(\gamma_k) := t''_{\mathcal{R}_k}(0) = \frac{f''(\gamma_k)}{f'(\gamma_k)}, \tag{1.6}$$

matching for $f=\Xi$ the Csordas–Smith–Varga zero-velocity ratio [3]. Moreover, for $f=\Xi$ in the Euler-product half-plane $d>\frac{1}{2}$ ($\sigma = \frac{1}{2}+d>1$, outside the critical strip), let $\mathcal{A}(d,t) := \mathrm{Re}(\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\ln\pi + \frac{1}{2}\psi(\frac{s}{2}))$. For every $t \ge 10$, the equation

$\mathcal{A}(d,t) = -\frac{\zeta'(1/2+d)}{\zeta(1/2+d)}$ *has a unique root* $d_{\mathrm{Euler}}(t) > \frac{1}{2}$*, and for all* $d > d_{\mathrm{Euler}}(t)$ *one has unconditionally* $\frac{W(d,t)}{|\Xi(t-\mathrm{i}d)|^2} = \partial_t \arg \Xi(t-\mathrm{i}d) \geq \mathcal{A}(d,t) + \frac{\zeta'(1/2+d)}{\zeta(1/2+d)} > 0$.

(iii) ***Transverse integral identity, conditional Rolle turning point, and hyperbolic low-pass filter:*** *For every real-on-*$\mathbb{R}$ *entire function* f *and all* $(d,t) \in \mathbb{R}^2$, $|f(t-\mathrm{i}d)|^2 = |f(t)|^2 + 2\int_0^d u\,\widetilde{W}_f(u,t)\,\mathrm{d}u$*; any zero* (d_*, t_*) *with* $d_* > 0$ *and* $L_1(f;t_*) > 0$ *forces an interior Rolle turning point* $d_{\mathrm{peak}} \in (0, d_*)$ *with* $\widetilde{W}_f(d_{\mathrm{peak}}, t_*) = 0$ *and* $\partial_d \widetilde{W}_f(d_{\mathrm{peak}}, t_*) \leq 0$*. For* $f = \Xi$ *under* (1.3)*, the normalized integral operator* $\frac{2}{d^2}\int_0^d u\,(\cdot)\,\mathrm{d}u$ *maps* $\mathrm{sinc}_{\mathrm{hyp}}(2uy)$ *to* $\mathrm{sinc}^2_{\mathrm{hyp}}(dy) = (\frac{\sinh(dy)}{dy})^2$*, multiplying each Wigner shift* $y > 0$ *by the contractive ratio*

$$\mathcal{F}_{\mathrm{int}}(2dy) := \frac{\mathrm{sinc}^2_{\mathrm{hyp}}(dy)}{\mathrm{sinc}_{\mathrm{hyp}}(2dy)} = \frac{\tanh(dy)}{dy} \in (0,1) \qquad (\forall\, dy > 0). \tag{1.7}$$

Representative numerical values of the local curvature, critical-line extrema, and Euler–Stirling boundary invariants for $f = \Xi$ established in Sections 3–5 are recorded below:

$$\begin{aligned} \kappa(\gamma_1) = \frac{\Xi''(\gamma_1)}{\Xi'(\gamma_1)} \approx -1.1592611550, \qquad c_{\mathrm{par}}(\gamma_1) = \frac{1}{2}\kappa(\gamma_1) \approx -0.5796305775, \\ \tau_1 \approx 15.5857085898, \qquad \kappa_\Gamma(\tau_1) = \frac{\Xi'''(\tau_1)}{3\,\Xi''(\tau_1)} \approx -0.37992197, \\ d_{\mathrm{Euler}}(\gamma_1) \approx 1.703275, \qquad d_{\mathrm{Euler}}(90) \approx 1.049797, \qquad d_{\mathrm{Euler}}(10^8) \approx 0.613019 \to \tfrac{1}{2}^+. \end{aligned} \tag{1.8}$$

Table 1 situates the exact identities and conditional statements of the present note relative to prior work.

TABLE 1. Relationship between classical prior art and the geometric and integral identities analyzed in this note.

Prior Literature	**Established Classical Result**	**Formulation in This Note**
Speiser (1934) [20], Lagarias (1999) [10], Arias de Reyna (2003) [1]	RH $\iff \zeta'(s) \neq 0$ in $0 < \sigma < \frac{1}{2}$ and $\iff \Xi'(t-\mathrm{i}d) \neq 0$ in $d > 0$; X-ray zero loci of ζ and ξ.	Arc-length identity $\frac{\mathrm{d}\lvert R\rvert}{\mathrm{d}s} = \pm\lvert\nabla I\rvert$ on $\{I = 0\}$, explicit two-mechanism dichotomy for off-critical zeros, and tracking to $d_{\mathrm{Euler}}(t) > \frac{1}{2}$.
Lehmer (1956) [11], Csordas–Norfolk–Varga (1986) [2], Csordas–Varga (1990) [4], Csordas–Smith–Varga (1994) [3]	First Laguerre expression $L_1(\Xi;t) = (\Xi')^2 - \Xi\Xi''$; zero-velocity ratio $\Xi''(\gamma_k)/\Xi'(\gamma_k)$ at Lehmer pairs.	Identification of $\Xi''(\gamma_k)/\Xi'(\gamma_k)$ as the nodal vertex curvature $t''_{\mathcal{R}_k}(0)$, fourth-order jet $c_{\mathrm{quart}}(\gamma_k)$, and rigorous `mpmath.iv` enclosures.
Hinkkanen (1997) [8], Spira (1965–1973) [21, 22, 23]	Pointwise criterion $\mathrm{Re}(\xi'(s)/\xi(s)) = W/\lvert\Xi\rvert^2 > 0$ in $d > 0$; zero-free regions of $\zeta^{(k)}(s)$ for σ large.	Transverse integral identity $\lvert f(t-\mathrm{i}d)\rvert^2 = \lvert f(t)\rvert^2 + 2\int_0^d u\widetilde{W}_f\,\mathrm{d}u$, conditional Rolle turning point, and explicit boundary $d_{\mathrm{Euler}}(t)$ in $\sigma > 1$.
Wigner (1932) [27], Pólya (1926–1927) [16, 17], Paper II [13]	Oscillatory Fourier kernels of $\Xi(t)$ and parity-reduced Wronskian $\widetilde{W}(d,t)$ with kernel $\mathrm{sinc}_{\mathrm{hyp}}(2dy)$.	Exact Fubini mapping $\mathrm{sinc}_{\mathrm{hyp}}(2uy) \mapsto \mathrm{sinc}^2_{\mathrm{hyp}}(dy)$ with hyperbolic filter ratio $\frac{\tanh(dy)}{dy} \in (0,1)$, alongside sign-oscillation obstructions.

1.1. **Proof overview.** The proof of Theorem 1.1 rests on three elementary complex-analytic and integral steps:

(1) *Harmonic boundary structure and Euler-product majorant in* $\sigma > 1$ *(Section 2):* For any entire function $f(t-\mathrm{i}d) = R_f(d,t) + \mathrm{i}I_f(d,t)$ real on $\mathbb{R}$, Schwarz reflection $f(t+\mathrm{i}d) = \overline{f(t-\mathrm{i}d)}$ and the Cauchy–Riemann system $(\partial_d R_f = \partial_t I_f,\ \partial_d I_f = -\partial_t R_f)$ yield

$$I_f(0,t) \equiv 0, \qquad \frac{\partial R_f}{\partial d}(0,t) \equiv 0, \qquad \frac{\partial I_f}{\partial d}(0,t) = -f'(t), \qquad \Delta R_f = \Delta I_f = 0. \tag{1.9}$$

Hence branches of $I_f(d,t)/d = 0$ meet $d = 0$ exclusively at critical points $f'(\tau) = 0$, the harmonic maximum principle precludes compact loops of $I_f = 0$, and for $f = \Xi$ in $\sigma = \frac{1}{2}+d > 1$ the triangle inequality $|\operatorname{Re}(\zeta'(s)/\zeta(s))| \le -\zeta'(\sigma)/\zeta(\sigma)$ enforces $\partial_t \arg \Xi > 0$ beyond $d_{\mathrm{Euler}}(t) > \frac{1}{2}$.

(2) *Arc-length differentiation along $\{I_f = 0\}$ and implicit zero-curve jets (Section 3):* Along a smooth curve $I_f(d(s), t(s)) \equiv 0$, ∇I_f is normal and ∇R_f is tangent with $|\nabla R_f| = |\nabla I_f| = |f'|$, giving (1.5) and the two-mechanism dichotomy (Proposition 3.2). Implicit differentiation of $R_f(d, t_{\mathcal{R}_k}(d)) = 0$ at a simple real zero γ_k using $R_{f,dd}(0,\gamma_k) = -R_{f,tt}(0,\gamma_k) = -f''(\gamma_k)$ gives

$$t'_{\mathcal{R}_k}(0) = 0, \qquad \kappa(\gamma_k) = t''_{\mathcal{R}_k}(0) = -\frac{R_{f,dd}(0,\gamma_k)}{R_{f,t}(0,\gamma_k)} = \frac{f''(\gamma_k)}{f'(\gamma_k)}. \tag{1.10}$$

(3) *Transverse integration and Wigner kernel identity (Section 4):* Integrating $\partial_u |f(t - \mathrm{i}u)|^2 = 2u\,\widetilde{W}_f(u,t)$ over $u \in [0,d]$ gives the transverse integral identity, while for $f = \Xi$ Fubini's theorem and $\frac{2}{d^2}\int_0^d u \operatorname{sinc}_{\mathrm{hyp}}(2uy)\,\mathrm{d}u = \operatorname{sinc}^2_{\mathrm{hyp}}(dy)$ yield

$$\begin{aligned} \widetilde{\mathcal{E}}(d,t) &:= \frac{|\Xi(t-\mathrm{i}d)|^2}{d^2} = \frac{\Xi(t)^2}{d^2} + 32\int_0^\infty y^2 \operatorname{sinc}^2_{\mathrm{hyp}}(dy)\, I_{\mathrm{Wig}}(y,t)\,\mathrm{d}y, \\ \mathcal{F}_{\mathrm{int}}(2dy) &:= \frac{\operatorname{sinc}^2_{\mathrm{hyp}}(dy)}{\operatorname{sinc}_{\mathrm{hyp}}(2dy)} = \frac{\tanh(dy)}{dy} \in (0,1). \end{aligned} \tag{1.11}$$

1.2. **Notation and conventions.** Throughout this paper, $s = \sigma + \mathrm{i}t = \frac{1}{2} + d + \mathrm{i}t$ with transverse offset $d = \sigma - \frac{1}{2} \in \mathbb{R}$ and spectral height $t \in \mathbb{R}$, and $z = -\mathrm{i}(s - \frac{1}{2}) = t - \mathrm{i}d$. For $\Xi(t - \mathrm{i}d) = \xi(\frac{1}{2} + d + \mathrm{i}t)$, we write $R(d,t) := \operatorname{Re}\Xi(t - \mathrm{i}d)$ and $I(d,t) := \operatorname{Im}\Xi(t - \mathrm{i}d)$; for a general entire function $f(z)$ real on $\mathbb{R}$, we write $f(t - \mathrm{i}d) = R_f(d,t) + \mathrm{i}I_f(d,t)$, $W_f(d,t) := \frac{1}{2}\partial_d |f(t - \mathrm{i}d)|^2$, $\widetilde{W}_f(d,t) := W_f(d,t)/d$, and $L_1(f;t) := (f'(t))^2 - f(t)f''(t) = \widetilde{W}_f(0,t)$ for the classical first Laguerre expression [9, 4]. To distinguish the imaginary part $I(d,t)$ from the Wigner correlation integral of $\Phi(u)$ in (1.3)–(1.4), the latter is written $I_{\mathrm{Wig}}(y,t)$. On $d = 0$, $\gamma_1 < \gamma_2 < \dots$ denote the positive zeros of $\Xi(t)$, and on any verified interval where consecutive zeros are simple and $L_1(\Xi;t) > 0$, $\tau_k \in (\gamma_k, \gamma_{k+1})$ denotes the unique critical-line extremum solving $\Xi'(\tau_k) = 0$ (distinct from the extrema of Hardy's function $Z(t)$; in particular $\tau_1 \approx 15.58570859$ and $\tau_2 \approx 22.09797728$). We write $E_0(t) := \frac{1}{2}(t^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \mathrm{i}\frac{t}{2})| > 0$ for the critical-line Archimedean envelope and indicate envelope-scaled quantities by hats as specified in context. Implicit constants in $\mathcal{O}(\cdot)$ and $\ll$ are absolute unless noted otherwise.

2. Harmonic nodal structure and the Euler–Stirling boundary in $\sigma > 1$

Let $f(z)$ be any non-constant entire function satisfying $f(\mathbb{R}) \subset \mathbb{R}$. Writing $f(t - \mathrm{i}d) = R_f(d,t) + \mathrm{i}I_f(d,t)$ on $\mathbb{R}^2$, complex differentiability with respect to $z = t - \mathrm{i}d$ implies the Cauchy–Riemann equations and harmonicity everywhere on $\mathbb{R}^2$:

$$\frac{\partial R_f}{\partial d} = \frac{\partial I_f}{\partial t}, \qquad \frac{\partial I_f}{\partial d} = -\frac{\partial R_f}{\partial t}, \qquad \Delta R_f := \frac{\partial^2 R_f}{\partial d^2} + \frac{\partial^2 R_f}{\partial t^2} = 0, \qquad \Delta I_f := \frac{\partial^2 I_f}{\partial d^2} + \frac{\partial^2 I_f}{\partial t^2} = 0. \tag{2.1}$$

We first record the exact boundary conditions on $d = 0$ and the local bifurcation of imaginary nodal curves, noting explicitly that this structure requires only $f(\mathbb{R}) \subset \mathbb{R}$.

Proposition 2.1 (Critical-line boundary conditions and orthogonal departure of $\{I_f = 0\}$)**.** *Let $f(z)$ be an entire function real on $\mathbb{R}$, and write $f(t - \mathrm{i}d) = R_f(d,t) + \mathrm{i}I_f(d,t)$. Then for all $t \in \mathbb{R}$:*

(a) ***Dirichlet nodal line, Neumann mirror condition, and zero Green flux:*** *Along $d = 0$, $I_f(0,t) \equiv 0$, $\partial_d R_f(0,t) \equiv 0$, and for every interval $[t_1, t_2] \subset \mathbb{R}$,*

$$\int_{t_1}^{t_2} R_f(0,t)\,\frac{\partial R_f}{\partial d}(0,t)\,\mathrm{d}t \equiv 0. \tag{2.2}$$

(b) ***Orthogonal departure of imaginary nodal walls from real critical points:*** *The parity-reduced imaginary part* $\widetilde{I}_f(d,t) := I_f(d,t)/d$ *is an even real-analytic function of* d *with Taylor expansion*

$$\widetilde{I}_f(d,t) = \sum_{m=0}^{\infty} \frac{(-1)^{m+1}}{(2m+1)!}\, f^{(2m+1)}(t)\, d^{2m} = -f'(t) + \frac{1}{6} f'''(t)\, d^2 - \frac{1}{120} f^{(5)}(t)\, d^4 + \mathcal{O}(d^6). \tag{2.3}$$

Hence a transverse nodal curve $\Gamma \subset \{I_f(d,t) = 0,\, d > 0\}$ *can meet* $d = 0$ *at* $(0,\tau)$ *if and only if* $f'(\tau) = 0$. *At every non-degenerate real critical point* τ ($f'(\tau) = 0$, $f''(\tau) \neq 0$), *the implicit function theorem yields a unique smooth curve* $t = t_\Gamma(d)$ *solving* $\widetilde{I}_f(d, t_\Gamma(d)) = 0$ *near* $d = 0$, *departing orthogonally with expansion*

$$t_\Gamma(d) - \tau = \frac{f'''(\tau)}{6\, f''(\tau)}\, d^2 + \mathcal{O}(d^4), \qquad t_\Gamma'(0) = 0, \qquad \kappa_\Gamma(\tau) := t_\Gamma''(0) = \frac{f'''(\tau)}{3\, f''(\tau)}. \tag{2.4}$$

Proof. (a) Since f is real on $\mathbb{R}$, Schwarz reflection gives $f(t+\mathrm{i}d) = \overline{f(t-\mathrm{i}d)}$, whence $R_f(-d,t) = R_f(d,t)$ is even in d and $I_f(-d,t) = -I_f(d,t)$ is odd in d. Setting $d = 0$ yields $I_f(0,t) \equiv 0$ and $\partial_d R_f(0,t) \equiv 0$, from which (2.2) follows immediately.

(b) Expanding $f(t - \mathrm{i}d) = \sum_{n=0}^{\infty} \frac{(-\mathrm{i}d)^n}{n!} f^{(n)}(t)$ into real and imaginary parts gives

$$R_f(d,t) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!}\, f^{(2m)}(t)\, d^{2m}, \qquad I_f(d,t) = d \sum_{m=0}^{\infty} \frac{(-1)^{m+1}}{(2m+1)!}\, f^{(2m+1)}(t)\, d^{2m}, \tag{2.5}$$

proving (2.3). For $d > 0$, $I_f(d,t) = 0 \iff \widetilde{I}_f(d,t) = 0$. If $(d_n, t_n) \in \{I_f = 0,\, d > 0\}$ converges to $(0,\tau)$, continuity gives $0 = \widetilde{I}_f(0,\tau) = -f'(\tau)$. Conversely, if $f'(\tau) = 0$ and $f''(\tau) \neq 0$, then $\widetilde{I}_f(0,\tau) = 0$, $\partial_d \widetilde{I}_f(0,\tau) = 0$, and $\partial_t \widetilde{I}_f(0,\tau) = -f''(\tau) \neq 0$. By the real-analytic implicit function theorem, $\widetilde{I}_f(d,t) = 0$ defines a unique even real-analytic curve $t = t_\Gamma(d)$ near $d = 0$ with $t_\Gamma(0) = \tau$ and $t_\Gamma'(0) = 0$. Substituting $t = \tau + \frac{1}{2}\kappa_\Gamma(\tau) d^2 + \mathcal{O}(d^4)$ into (2.3) gives $-f''(\tau) \cdot \frac{1}{2}\kappa_\Gamma(\tau) d^2 + \frac{1}{6} f'''(\tau) d^2 + \mathcal{O}(d^4) = 0$, proving (2.4). □

Next we separate the universal harmonic maximum-principle consequences from the arithmetic Euler-product bound for $\Xi(t - \mathrm{i}d)$, which holds exclusively in the half-plane $d > \frac{1}{2}$ ($\sigma > 1$, outside the critical strip).

Proposition 2.2 (Compact nodal loops and Euler–Stirling boundary)**.** (a) ***Exclusion of compact loops and chamber sign:*** *Let* $f(z)$ *be any non-constant entire function real on* $\mathbb{R}$, *with* $f(t - \mathrm{i}d) = R_f + \mathrm{i}I_f$. *No non-empty bounded open set* $U \subset \mathbb{R}^2$ *can satisfy* $I_f = 0$ *on* ∂U; *hence no branch of* $I_f(d,t) = 0$ *forms a closed loop in* $d > 0$ *or returns to* $d = 0$ *to enclose a bounded lens. On any real interval* $[T_1, T_2]$ *where the zeros* γ_k *of* f *are simple and* $L_1(f;t) > 0$, *each* (γ_k, γ_{k+1}) *contains a unique critical point* τ_k *with* $f'(\tau_k) = 0$ *and* $f''(\tau_k) \neq 0$, *and on each component* Ω_k *of* $\{d > 0 : I_f \neq 0\}$ *meeting* $d = 0$ *along* (τ_{k-1}, τ_k) *one has* $I_f \neq 0$ *(so* $f \neq 0$*). Provided the bounding walls* Γ_{k-1}, Γ_k *meet no critical point* $f'(z) = 0$ *in* $0 < d \leq d_0$, *they remain smooth and unbranched on* $0 < d \leq d_0$.

(b) ***Euler–Stirling right boundary*** $d_{\mathrm{Euler}}(t) > \frac{1}{2}$ ***for*** Ξ ***in*** $\sigma > 1$***:*** *Specialize to* $f = \Xi$. *For* $d > \frac{1}{2}$ ($\sigma = \frac{1}{2} + d > 1$, *outside the critical strip*) *and* $t > 0$, *define*

$$\begin{aligned} \mathcal{A}(d,t) &:= \mathrm{Re}\left(\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\ln\pi + \frac{1}{2}\psi\left(\frac{s}{2}\right)\right), \\ M_{\mathrm{Euler}}(d) &:= \sum_{n=2}^{\infty} \frac{\Lambda(n)}{n^{1/2+d}} = -\frac{\zeta'\left(\frac{1}{2}+d\right)}{\zeta\left(\frac{1}{2}+d\right)} > 0. \end{aligned} \tag{2.6}$$

For every $t \geq 10$, *the equation* $\mathcal{A}(d,t) = M_{\mathrm{Euler}}(d)$ *admits a unique root* $d = d_{\mathrm{Euler}}(t) > \frac{1}{2}$, *and for all* $d > d_{\mathrm{Euler}}(t)$ *one has unconditionally*

$$\frac{W(d,t)}{\left|\Xi(t - \mathrm{i}d)\right|^2} = \frac{\partial}{\partial t} \arg \Xi(t - \mathrm{i}d) \geq \mathcal{A}(d,t) - M_{\mathrm{Euler}}(d) = \mathcal{A}(d,t) + \frac{\zeta'\left(\frac{1}{2}+d\right)}{\zeta\left(\frac{1}{2}+d\right)} > 0. \tag{2.7}$$

Along any vertical line $d = d_0 > \sup_{t\in[t_a,t_b]} d_{\mathrm{Euler}}(t)$, $\arg \Xi(t - \mathrm{i}d_0)$ increases strictly with t. As $t \to \infty$, $d_{\mathrm{Euler}}(t) = \frac{1}{2} + [\frac{1}{2} \ln \frac{t}{2\pi} + \gamma_{\mathrm{E}} + \mathcal{O}((\ln t)^{-1})]^{-1} \to \frac{1}{2}^+$ from the right of $\sigma = 1$.

Proof. (a) If $U \subset \mathbb{R}^2$ is a non-empty bounded open domain with $I_f|_{\partial U} \equiv 0$, harmonicity $\Delta I_f = 0$ and the maximum and minimum principles give $0 = \min_{\partial U} I_f \leq I_f \leq \max_{\partial U} I_f = 0$ on U. Real-analyticity then forces $I_f \equiv 0$ on $\mathbb{R}^2$, whence $\partial_t R_f = -\partial_d I_f \equiv 0$ and $\partial_d R_f = \partial_t I_f \equiv 0$, contradicting that f is non-constant. On any real interval where all zeros γ_k of f are simple and $L_1(f;t) > 0$, every critical point $f'(\tau) = 0$ satisfies $-f(\tau)f''(\tau) = L_1(f;\tau) > 0$, so f is strictly concave where $f > 0$ and strictly convex where $f < 0$; Rolle's theorem then gives a unique critical point $\tau_k \in (\gamma_k, \gamma_{k+1})$. By connectivity, I_f has constant non-zero sign inside each component Ω_k of $\{d > 0 : I_f \neq 0\}$, and by the implicit function theorem Γ_{k-1}, Γ_k cannot split unless $\nabla I_f = (-\partial_t R_f, \partial_d R_f) = (0, 0)$, i.e. $f'(z) = 0$.

(b) For $\sigma = \frac{1}{2} + d > 1$, the Euler product of $\zeta(s)$ converges absolutely, so $\zeta(s) \neq 0$ and $-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=2}^{\infty} \Lambda(n) n^{-1/2-d-\mathrm{i}t}$. By the triangle inequality, $|\operatorname{Re}(\zeta'(s)/\zeta(s))| \leq \sum_{n=2}^{\infty} \Lambda(n) n^{-1/2-d} = M_{\mathrm{Euler}}(d)$. Taking real parts of $\frac{\xi'(s)}{\xi(s)} = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\ln \pi + \frac{1}{2}\psi(\frac{s}{2}) + \frac{\zeta'(s)}{\zeta(s)}$ and noting $\operatorname{Re}(\frac{\xi'(s)}{\xi(s)}) = \frac{W(d,t)}{|\Xi(t-\mathrm{i}d)|^2} = \partial_t \arg \Xi(t - \mathrm{i}d)$ yields (2.7). On $d > \frac{1}{2}$ with $t \geq 10$ fixed, $M_{\mathrm{Euler}}(d)$ decreases strictly from $+\infty$ to 0 while $\mathcal{A}(d,t)$ increases strictly with d, ensuring a unique root $d_{\mathrm{Euler}}(t) > \frac{1}{2}$; its asymptotic expansion is proved in Lemma A.1(b). □

3. Cauchy–Riemann arc-length law, off-critical dichotomy, and zero-curve curvature

Because $f(t - \mathrm{i}d) = R_f(d,t) + \mathrm{i}I_f(d,t)$ cannot vanish where $I_f(d,t) \neq 0$, every off-critical zero (d_*, t_*) $(d_* > 0)$ must lie on the imaginary nodal locus $\{I_f(d,t) = 0\}$. We first state the universal Cauchy–Riemann arc-length equation along $\{I_f = 0\}$ and then prove the exact two-mechanism dichotomy showing why global monotonicity cannot hold unconditionally without RH-equivalent hypotheses.

Theorem 3.1 (Universal Cauchy–Riemann arc-length identity along $\{I_f = 0\}$)**.** *Let $f(z)$ be an entire function real on $\mathbb{R}$, write $f(t-\mathrm{i}d) = R_f(d,t)+\mathrm{i}I_f(d,t)$, and let $\Gamma = \{(d(s), t(s)) : s \in [0, S]\}$ be a smooth arc-length parameterized segment ($d'(s)^2 + t'(s)^2 = 1$) of $\{I_f(d,t) = 0\}$ departing orthogonally into $d > 0$ from a real critical point $(0,\tau)$ where $f'(\tau) = 0$, $f(\tau) \neq 0$, and $f''(\tau) \neq 0$. Assume $f'(t(s) - \mathrm{i}d(s)) \neq 0$ and $R_f(d(s), t(s)) \neq 0$ for $s \in (0, S]$. Then:*

(a) *Along Γ, ∇R_f is tangent to Γ and ∇I_f is normal to Γ, with $|\nabla R_f| = |\nabla I_f| = |f'(t-\mathrm{i}d)|$.*

(b) *Setting $\varepsilon(\tau) := \operatorname{sgn}(L_1(f;\tau)) = \operatorname{sgn}(-f(\tau)f''(\tau)) \in \{+1, -1\}$, the modulus $|R_f(d(s), t(s))|$ equals $|f(t(s) - \mathrm{i}d(s))|$ and satisfies*

$$\frac{\mathrm{d}}{\mathrm{d}s}\big|R_f(d(s),t(s))\big| = \varepsilon(\tau)\,\big|\nabla I_f(d(s),t(s))\big| = \varepsilon(\tau)\,\big|f'(t(s) - \mathrm{i}d(s))\big| \qquad (\forall\, s \in (0, S]). \tag{3.1}$$

In particular, if $L_1(f;\tau) > 0$ and $f'(t(s)-\mathrm{i}d(s)) \neq 0$ for all $s \in (0, S]$, then $|R_f|$ increases strictly:

$$|R_f(d(s),t(s))| = |f(\tau)| + \int_0^s \big|f'(t(u) - \mathrm{i}d(u))\big|\,\mathrm{d}u > |f(\tau)| > 0 \qquad (\forall\, s \in (0, S]). \tag{3.2}$$

Proof. Differentiating $I_f(d(s), t(s)) = 0$ with respect to arc length s gives $\nabla I_f \cdot \mathbf{T}(s) = 0$ for the unit tangent $\mathbf{T}(s) = (d'(s), t'(s))$. By (2.1),

$$\nabla R_f \cdot \nabla I_f = \frac{\partial R_f}{\partial d}\left(-\frac{\partial R_f}{\partial t}\right) + \frac{\partial R_f}{\partial t}\frac{\partial R_f}{\partial d} \equiv 0, \qquad |\nabla R_f| = |\nabla I_f| = |f'(t - \mathrm{i}d)|. \tag{3.3}$$

Thus ∇R_f is parallel to $\mathbf{T}(s)$, whence $|\frac{\mathrm{d}}{\mathrm{d}s} R_f(d(s), t(s))| = |\nabla R_f \cdot \mathbf{T}(s)| = |\nabla R_f| = |\nabla I_f| = |f'(t(s) - \mathrm{i}d(s))|$. Near $d = 0$, Proposition 2.1(b) gives $s = d + \mathcal{O}(d^3)$ and $t_\Gamma(d) - \tau = \mathcal{O}(d^2)$, so (2.5) with $f'(\tau) = 0$ yields

$$f(\tau)\, R_f(d, t_\Gamma(d)) = f(\tau)^2 + \frac{1}{2}(-f(\tau)f''(\tau))\, d^2 + \mathcal{O}(d^4) = f(\tau)^2 + \frac{1}{2}L_1(f;\tau)\, d^2 + \mathcal{O}(d^4).$$

Hence the initial sign of $\frac{\mathrm{d}|R_f|}{\mathrm{d}s}$ for small $s > 0$ equals $\varepsilon(\tau) = \operatorname{sgn}(L_1(f;\tau))$. Since $|f'(t(s)-\mathrm{i}d(s))| > 0$ and $R_f(d(s),t(s)) \neq 0$ on $(0,S]$, continuity of $\frac{\mathrm{d}|R_f|}{\mathrm{d}s}$ preserves this sign throughout $(0,S]$, proving (3.1) and, when $\varepsilon(\tau) = +1$, (3.2). □

Proposition 3.2 (Two-mechanism dichotomy for off-critical zeros on nodal walls)**.** *Let $f(z)$ be a non-constant entire function real on $\mathbb{R}$, and suppose $f(t_* - \mathrm{i}d_*) = 0$ at an off-critical point (d_*, t_*) with $d_* > 0$ lying on a connected nodal branch $\Gamma \subset \{I_f(d,t) = 0,\ d \geq 0\}$ that meets $d = 0$ at a real point $(0,\tau)$. Then at least one of the following two obstructions must occur:*

(i) ***Boundary Laguerre failure on** $d = 0$**:** *$L_1(f;\tau) = (f'(\tau))^2 - f(\tau)f''(\tau) \leq 0$ at $(0,\tau)$ (so either τ is a multiple real zero $f(\tau) = f'(\tau) = 0$, a degenerate critical point $f'(\tau) = f''(\tau) = 0$, or a wrong-concavity extremum $-f(\tau)f''(\tau) < 0$ from which $|R_f|$ decreases with $\frac{\mathrm{d}|R_f|}{\mathrm{d}s} = -|f'| < 0$ directly to 0 at (d_*,t_*) without any interior critical point in $d > 0$); or*

(ii) ***Interior critical point in** $d > 0$**:** *$L_1(f;\tau) > 0$ holds at $(0,\tau)$, and Γ passes through an interior critical point $z_{\mathrm{crit}} = t_{\mathrm{crit}} - \mathrm{i}d_{\mathrm{crit}}$ ($d_{\mathrm{crit}} > 0$, $f'(z_{\mathrm{crit}}) = 0$) where two or more nodal branches of $I_f(d,t) = 0$ intersect and $\frac{\mathrm{d}|R_f|}{\mathrm{d}s}$ reverses sign.*

Both mechanisms are realized explicitly by the holomorphic model family $\Xi_{\mathrm{hyp}}(z;d_0) := \Xi(z)(1 + (z - 17.85)^2/d_0^2)$: for $d_0 = 0.25$ one obtains mechanism (i) *at $\tau_* \approx 17.87156653$ with $L_1(\Xi_{\mathrm{hyp}};\tau_*) < 0$ and no critical point in $d > 0$, whereas for $d_0 = 1.30$ one obtains mechanism* (ii) *at the interior saddle $(d_{\mathrm{crit}}, t_{\mathrm{crit}}) \approx (0.56054034, 18.66151713)$ while $L_1 > 0$ holds at the real extrema.*

Proof. Assume toward a contradiction that neither (i) nor (ii) occurs along the path in Γ joining $(0,\tau)$ to (d_*,t_*). Since $(0,\tau)$ is a limit point of $\{I_f = 0,\ d > 0\}$, Proposition 2.1(b) gives $f'(\tau) = 0$, and negation of (i) gives $L_1(f;\tau) = -f(\tau)f''(\tau) > 0$, so $f(\tau) \neq 0$ and $f''(\tau) \neq 0$. Negation of (ii) implies $f'(z) \neq 0$ along the entire arc from $(0,\tau)$ to (d_*,t_*), so the arc is a single smooth curve parameterized by arc length $s \in [0,s_*]$ with $(d(s_*),t(s_*)) = (d_*,t_*)$. Let $s_0 \in (0,s_*]$ be the first arc-length parameter at which $R_f(d(s),t(s)) = 0$. Applying Theorem 3.1(b) on $[0, s_0 - \delta]$ and letting $\delta \to 0^+$ yields $|R_f(d(s_0),t(s_0))| = |f(\tau)| + \int_0^{s_0} |f'(t(u) - \mathrm{i}d(u))|\,\mathrm{d}u > |f(\tau)| > 0$, contradicting $R_f(d(s_0),t(s_0)) = 0$. The two explicit instances for $\Xi_{\mathrm{hyp}}(z;d_0)$ at $d_0 = 0.25$ and $d_0 = 1.30$ are verified to 50 decimal digits in Section 5.3. □

Remark 3.3 (Relation to Lagarias's positivity theorem and Speiser's theorem). Proposition 3.2 makes transparent why Theorem 3.1 is a geometric reformulation rather than an unconditional bypass of RH: for $f = \Xi$, Lagarias [10, Theorem 1.1] proved via Hadamard factorization that $\mathrm{Re}(\xi'(s)/\xi(s)) > 0$ (and hence $\Xi'(t - \mathrm{i}d) \neq 0$) on $d > 0$ is equivalent to RH, while $L_1(\Xi;t) > 0$ on $\mathbb{R}$ is the degree-2 Jensen–Csordas–Varga necessary condition [9, 4].

We next derive the local parabolic curvature and fourth-order Taylor jet of the real zero curves $\mathcal{R}_k := \{(d,t) : R_f(d,t) = 0\}$ departing from simple real zeros γ_k, connecting $\kappa(\gamma_k)$ to the Csordas–Smith–Varga velocity [3].

Theorem 3.4 (Zero-curve curvature $\kappa(\gamma_k)$ and fourth-order jet of $\{R_f(d,t) = 0\}$)**.** *Let $f(z)$ be an entire function real on $\mathbb{R}$, and let $\gamma_k \in \mathbb{R}$ be a simple zero ($f(\gamma_k) = 0$, $f'(\gamma_k) \neq 0$). In a neighborhood of $(0,\gamma_k)$, the real zero locus $R_f(d,t) = 0$ is a smooth curve $t = t_{\mathcal{R}_k}(d)$ departing orthogonally from $d = 0$ ($t'_{\mathcal{R}_k}(0) = 0$) with expansion*

$$t_{\mathcal{R}_k}(d) - \gamma_k = c_{\mathrm{par}}(\gamma_k)\, d^2 + c_{\mathrm{quart}}(\gamma_k)\, d^4 + \mathcal{O}(d^6), \tag{3.4}$$

where the vertex curvature $\kappa(\gamma_k) := t''_{\mathcal{R}_k}(0)$, parabolic coefficient $c_{\mathrm{par}}(\gamma_k) = \frac{1}{2}\kappa(\gamma_k)$, and quartic coefficient $c_{\mathrm{quart}}(\gamma_k) := \frac{1}{24}t^{(4)}_{\mathcal{R}_k}(0)$ are given by

$$\begin{aligned} \kappa(\gamma_k) &= \frac{f''(\gamma_k)}{f'(\gamma_k)}, \qquad c_{\mathrm{par}}(\gamma_k) = \frac{f''(\gamma_k)}{2\,f'(\gamma_k)}, \\ c_{\mathrm{quart}}(\gamma_k) &= \frac{12\,f'''(\gamma_k)\,c_{\mathrm{par}}(\gamma_k) - 12\,f''(\gamma_k)\,c_{\mathrm{par}}(\gamma_k)^2 - f^{(4)}(\gamma_k)}{24\,f'(\gamma_k)}. \end{aligned} \tag{3.5}$$

For $f = \Xi$, writing $\xi(s) = G(s)\zeta(s)$ with $G(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)$ and $g_1(s) = \frac{G'(s)}{G(s)}$, $\kappa(\gamma_k)$ decomposes at $s_k = \frac{1}{2} + \mathrm{i}\gamma_k$ as

$$\begin{aligned} \kappa(\gamma_k) &= -\operatorname{Im}\left(2\,g_1\left(\frac{1}{2}+\mathrm{i}\gamma_k\right) + \frac{\zeta''\left(\frac{1}{2}+\mathrm{i}\gamma_k\right)}{\zeta'\left(\frac{1}{2}+\mathrm{i}\gamma_k\right)}\right), \\ \operatorname{Re}\left(\frac{\zeta''\left(\frac{1}{2}+\mathrm{i}\gamma_k\right)}{\zeta'\left(\frac{1}{2}+\mathrm{i}\gamma_k\right)}\right) &= -2\,\operatorname{Re} g_1\left(\frac{1}{2}+\mathrm{i}\gamma_k\right). \end{aligned} \tag{3.6}$$

Proof. At $(0,\gamma_k)$, $R_f(0,\gamma_k) = f(\gamma_k) = 0$, $\partial_d R_f(0,\gamma_k) = 0$ by Proposition 2.1(a), and $\partial_t R_f(0,\gamma_k) = f'(\gamma_k) \neq 0$. Since $R_f(-d,t) = R_f(d,t)$ is even in d, the implicit function theorem gives a unique even real-analytic curve $t = t_{\mathcal{R}_k}(d)$ solving $R_f(d, t_{\mathcal{R}_k}(d)) = 0$ near $d = 0$, with $t_{\mathcal{R}_k}(0) = \gamma_k$ and $t_{\mathcal{R}_k}^{(2m+1)}(0) = 0$ for all $m \geq 0$. Differentiating $R_f(d,t(d)) = 0$ twice at $d = 0$ and using $t'(0) = 0$ and $R_{f,dd}(0,\gamma_k) = -R_{f,tt}(0,\gamma_k) = -f''(\gamma_k)$ gives

$$R_{f,dd}(0,\gamma_k) + R_{f,t}(0,\gamma_k)\,t''(0) = -f''(\gamma_k) + f'(\gamma_k)\,t''(0) = 0,$$

proving $\kappa(\gamma_k) = f''(\gamma_k)/f'(\gamma_k)$; the quartic formula is derived in Lemma A.1(a). For $f = \Xi$, differentiating $\Xi(t) = G(\frac{1}{2}+\mathrm{i}t)\zeta(\frac{1}{2}+\mathrm{i}t)$ at $s_k = \frac{1}{2}+\mathrm{i}\gamma_k$ ($\zeta(s_k) = 0$) gives $\Xi'(\gamma_k) = \mathrm{i}G(s_k)\zeta'(s_k)$ and $\Xi''(\gamma_k) = -G(s_k)(2g_1(s_k)\zeta'(s_k) + \zeta''(s_k))$, so $\kappa(\gamma_k) = \mathrm{i}(2g_1(s_k) + \zeta''(s_k)/\zeta'(s_k))$. Since $\kappa(\gamma_k) \in \mathbb{R}$, taking real and imaginary parts yields (3.6). □

Remark 3.5 (Connection to Csordas–Smith–Varga velocity and Lehmer pairs). Under the de Bruijn–Newman heat evolution H_Λ [6, 14, 19], Csordas, Smith, and Varga [3, Equation (2.8)] proved that a simple zero $x_k(\Lambda)$ moves with velocity $x_k'(0) = -\frac{H_0''(x_k)}{H_0'(x_k)}$, which up to scaling by 2 coincides with $-\kappa(\gamma_k) = -\Xi''(\gamma_k)/\Xi'(\gamma_k)$. Because $\Delta R = 0$ shares the second-derivative operator $\partial_d^2 = -\partial_t^2$ with backward heat continuation, closely spaced Lehmer pairs [11] where $|\Xi'(\gamma_k)|$ is small produce rapid zero motion under H_Λ and simultaneously sharp bending $|\kappa(\gamma_k)|$ of the nodal locus $R(d,t) = 0$ (e.g. $\kappa(\gamma_9) \approx -2.3970$ and $\kappa(\gamma_{13}) \approx -2.2519$ in Table 2).

4. Transverse integral identity, conditional Rolle turning point, and hyperbolic low-pass filter

We now record the exact transverse integral identity relating $|f(t-\mathrm{i}d)|^2$ to $\widetilde{W}_f(u,t)$ for any real-on-$\mathbb{R}$ entire function f, its conditional Rolle turning-point consequence at hypothetical off-critical zeros, and the Wigner kernel transformation for $f = \Xi$.

Theorem 4.1 (Transverse integral identity and conditional Rolle turning point)**.** *Let $f(z)$ be an entire function real on $\mathbb{R}$, with $\widetilde{W}_f(d,t) = \frac{1}{2d}\partial_d|f(t-\mathrm{i}d)|^2$. For all $(d,t) \in \mathbb{R}^2$,*

$$\boxed{\begin{aligned} |f(t-\mathrm{i}d)|^2 &= |f(t)|^2 + 2\int_0^d u\,\widetilde{W}_f(u,t)\,\mathrm{d}u \\ &= \left[f(t)^2 + d^2(f'(t))^2\right] - d^2 f(t)f''(t) + 2\int_0^d u(\widetilde{W}_f(u,t) - \widetilde{W}_f(0,t))\,\mathrm{d}u. \end{aligned}} \tag{4.1}$$

If (d_*,t_*) with $d_* > 0$ satisfies $f(t_* - \mathrm{i}d_*) = 0$, then:

(a) ***Integral identity at an off-critical zero:***

$$\int_0^{d_*} u\,\widetilde{W}_f(u,t_*)\,\mathrm{d}u = -\frac{1}{2}|f(t_*)|^2 \leq 0. \tag{4.2}$$

(b) ***Conditional Rolle turning point:*** *Under the hypothesis that $L_1(f;t_*) = \widetilde{W}_f(0,t_*) = (f'(t_*))^2 - f(t_*)f''(t_*) > 0$ (or at an m-fold real zero of f where $c_{2m-1}(f;t_*) = \frac{m}{(m!)^2}(f^{(m)}(t_*))^2 > 0$), the profile $E(u) := |f(t_* - \mathrm{i}u)|^2$ attains a strict interior maximum*

at some $d_{\mathrm{peak}} \in (0, d_*)$ *satisfying*

$$\widetilde{W}_f(d_{\mathrm{peak}}, t_*) = 0, \qquad \frac{\partial \widetilde{W}_f}{\partial d}(d_{\mathrm{peak}}, t_*) \le 0, \qquad \int_{d_{\mathrm{peak}}}^{d_*} 2u\, \widetilde{W}_f(u, t_*)\, \mathrm{d}u = -\big|f(t_* - \mathrm{i}d_{\mathrm{peak}})\big|^2 < 0. \tag{4.3}$$

Proof. Since $\partial_u |f(t - \mathrm{i}u)|^2 = 2W_f(u,t) = 2u\,\widetilde{W}_f(u,t)$, integrating over $u \in [0,d]$ yields the first line of (4.1). Adding and subtracting $\widetilde{W}_f(0,t) = L_1(f;t) = (f'(t))^2 - f(t)f''(t)$ inside the integral and evaluating $\int_0^d 2u\,\widetilde{W}_f(0,t)\,\mathrm{d}u = d^2(f'(t))^2 - d^2 f(t)f''(t)$ yields the second line. Setting $(d,t) = (d_*, t_*)$ when $f(t_* - \mathrm{i}d_*) = 0$ gives (4.2). Under the hypothesis $L_1(f;t_*) > 0$ (or $c_{2m-1}(f;t_*) > 0$), $E(0) \ge 0$, $E'(0) = 0$, and the first non-vanishing even derivative of $E(u)$ at $u = 0$ is strictly positive, so $E(u) > E(0) \ge 0 = E(d_*)$ for small $u > 0$. Compactness of $[0, d_*]$ yields an interior global maximum $d_{\mathrm{peak}} \in (0, d_*)$ with $E(d_{\mathrm{peak}}) > 0$, $E'(d_{\mathrm{peak}}) = 2d_{\mathrm{peak}}\widetilde{W}_f(d_{\mathrm{peak}}, t_*) = 0$, $E''(d_{\mathrm{peak}}) = 2d_{\mathrm{peak}}\partial_d \widetilde{W}_f(d_{\mathrm{peak}}, t_*) \le 0$, and $\int_{d_{\mathrm{peak}}}^{d_*} 2u\widetilde{W}_f(u,t_*)\,\mathrm{d}u = E(d_*) - E(d_{\mathrm{peak}}) = -E(d_{\mathrm{peak}}) < 0$. □

Specializing (4.1) to $f = \Xi$ under the canonical Wigner representation (1.3), we obtain the exact pure-square kernel identity and hyperbolic low-pass filter ratio.

Theorem 4.2 (Wigner $\mathrm{sinc}_{\mathrm{hyp}}^2$ kernel identity, hyperbolic low-pass filter, and baseline squares)**.** *For $f = \Xi$ and $d > 0$, define $\widetilde{\mathcal{E}}(d,t) := \frac{1}{d^2}|\Xi(t - \mathrm{i}d)|^2 = \frac{1}{d^2}|\Xi(t)|^2 + \frac{2}{d^2}\int_0^d u\,\widetilde{W}(u,t)\,\mathrm{d}u$.*

(a) ***Squared hyperbolic sinc kernel and low-pass ratio:*** *Under (1.3)–(1.4),*

$$\frac{2}{d^2}\int_0^d u\,\widetilde{W}(u,t)\,\mathrm{d}u = 32\int_0^\infty y^2\,\mathrm{sinc}_{\mathrm{hyp}}^2(dy)\,I_{\mathrm{Wig}}(y,t)\,\mathrm{d}y, \qquad \mathrm{sinc}_{\mathrm{hyp}}^2(dy) = \left(\frac{\sinh(dy)}{dy}\right)^2 \ge 1. \tag{4.4}$$

Relative to the pointwise kernel $\mathrm{sinc}_{\mathrm{hyp}}(2dy)$ in (1.3), the operator $\frac{2}{d^2}\int_0^d u\,(\cdot)\,\mathrm{d}u$ multiplies every Wigner shift $y > 0$ by the strictly decreasing ratio

$$\mathcal{F}_{\mathrm{int}}(2dy) := \frac{\frac{2}{d^2}\int_0^d u\,\mathrm{sinc}_{\mathrm{hyp}}(2uy)\,\mathrm{d}u}{\mathrm{sinc}_{\mathrm{hyp}}(2dy)} = \frac{\mathrm{sinc}_{\mathrm{hyp}}^2(dy)}{\mathrm{sinc}_{\mathrm{hyp}}(2dy)} = \frac{\tanh(dy)}{dy} \in (0,1) \qquad (\forall\, dy > 0), \tag{4.5}$$

with $\mathcal{F}_{\mathrm{int}}(0) = 1$, $\mathcal{F}_{\mathrm{int}}'(w) < 0$ for $w > 0$, and $\mathcal{F}_{\mathrm{int}}(2dy) \sim (dy)^{-1}$ as $dy \to \infty$.

(b) ***Baseline sum of squares*** *$\mathcal{Q}_0(d,t) := \Xi(t)^2 + d^2(\Xi'(t))^2$**:*** *On any real interval where all zeros of $\Xi(t)$ are simple, $\mathcal{Q}_0(d,t) > 0$ for every $d > 0$: at a simple zero γ_k, $\mathcal{Q}_0(d,\gamma_k) = d^2(\Xi'(\gamma_k))^2 = d^2 L_1(\Xi;\gamma_k) > 0$ while $-d^2\Xi(\gamma_k)\Xi''(\gamma_k) = 0$; at a real extremum τ_k ($\Xi'(\tau_k) = 0$) with $L_1(\Xi;\tau_k) > 0$, $\mathcal{Q}_0(d,\tau_k) = \Xi(\tau_k)^2 > 0$ while $-d^2\Xi(\tau_k)\Xi''(\tau_k) = d^2 L_1(\Xi;\tau_k) > 0$.*

Proof. (a) By [12, Theorem 1.1], $|I_{\mathrm{Wig}}(y,t)| \le I_{\mathrm{Wig}}(y,0) \ll \mathrm{e}^{8y}\exp(-2\pi\mathrm{e}^{2y})$ uniformly in $t \in \mathbb{R}$, so $64uy^2\,\mathrm{sinc}_{\mathrm{hyp}}(2uy)|I_{\mathrm{Wig}}(y,t)| \le 32y\sinh(2dy)I_{\mathrm{Wig}}(y,0) \in L^1([0,d]\times[0,\infty))$. Fubini's theorem therefore justifies interchanging the integrals, and $\cosh(2dy) - 1 = 2\sinh^2(dy)$ gives

$$\frac{2}{d^2}\int_0^d u\,\frac{\sinh(2uy)}{2uy}\,\mathrm{d}u = \frac{\cosh(2dy) - 1}{2d^2y^2} = \frac{\sinh^2(dy)}{(dy)^2} = \mathrm{sinc}_{\mathrm{hyp}}^2(dy),$$

proving (4.4). Using $\sinh(2dy) = 2\sinh(dy)\cosh(dy)$ gives $\frac{\mathrm{sinc}_{\mathrm{hyp}}^2(dy)}{\mathrm{sinc}_{\mathrm{hyp}}(2dy)} = \frac{\tanh(dy)}{dy}$, and $\frac{\mathrm{d}}{\mathrm{d}w}\big(\frac{\tanh w}{w}\big) = \frac{2w - \sinh(2w)}{2w^2\cosh^2 w} < 0$ for $w > 0$ establishes (4.5). Part (b) follows directly by evaluating (4.1) at γ_k and τ_k. □

5. Numerical and interval diagnostics

All local jets, nodal curves, interval enclosures, and counterexample checks in Sections 2–4 are reproduced by the standalone Python script `scripts/verify_paper3_harmonic_nodal_chambers.py`. As noted in Section 1, simplicity and critical-line location of all zeros on $t \in [0, 1000]$ are classical [7, 25, 15]; the computations below are finite-domain geometric diagnostics carried out at `mp.dps = 50` (and `iv.dps = 55` in `mpmath.iv` for Table 2).

5.1. **Zero-curve curvature $\kappa(\gamma_k)$ and rigorous interval enclosures across $\gamma_1,\dots,\gamma_{15}$.** Table 2 lists the scaled derivatives $\widehat{\Xi}'(\gamma_k) := \mathrm{e}^{\pi\gamma_k/4}\Xi'(\gamma_k)$ and $\widehat{\Xi}''(\gamma_k) := \mathrm{e}^{\pi\gamma_k/4}\Xi''(\gamma_k)$, the vertex curvature $\kappa(\gamma_k) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$, the parabolic and quartic coefficients $c_{\mathrm{par}}(\gamma_k)$ and $c_{\mathrm{quart}}(\gamma_k)$ from Theorem 3.4, and the discrepancy $|t_{\mathcal{R}_k}(0.05) - (\gamma_k + c_{\mathrm{par}}d^2 + c_{\mathrm{quart}}d^4)|$ at $d = 0.05$ across $\gamma_1,\dots,\gamma_{15}$. Across all 15 zeros, the decomposition (3.6) agrees with $\Xi''(\gamma_k)/\Xi'(\gamma_k)$ to within 1.28×10^{-49}. In addition, each γ_k ($1 \le k \le 15$) is enclosed rigorously in `mpmath.iv` (`iv.dps = 55`) via intermediate-value sign changes of $\Xi(\gamma_k \pm 10^{-22})$ using $N_u = 160$-panel composite trapezoidal quadrature of (1.1) on $u \in [0, 1.55]$ with proven analytic strip-discretization and tail bounds $< 10^{-65}$, confirming $\kappa(\gamma_k) \in \boldsymbol{\kappa}_k < 0$ with maximum interval width 3.27×10^{-5}. All 15 curvatures are negative due to the Archimedean digamma drift $-2\,\mathrm{Im}\,g_1(\frac{1}{2} + \mathrm{i}\gamma_k) \approx -\frac{\pi}{2} \approx -1.5708$.

TABLE 2. Zero-curve vertex curvature $\kappa(\gamma_k) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$, parabolic coefficient $c_{\mathrm{par}}(\gamma_k) = \frac{1}{2}\kappa(\gamma_k)$, quartic coefficient $c_{\mathrm{quart}}(\gamma_k)$, and tracking residual at $d = 0.05$ across $\gamma_1,\dots,\gamma_{15}$ (`mp.dps = 50`; negative sign enclosed rigorously in `mpmath.iv`).

Zero	**Height γ_k**	$\widehat{\Xi}'(\gamma_k)$	$\widehat{\Xi}''(\gamma_k)$	$\kappa(\gamma_k) = \Xi''/\Xi'$	$c_{\mathrm{par}}(\gamma_k) = \frac{1}{2}\kappa$	$c_{\mathrm{quart}}(\gamma_k)$	**Jet Error ($d = 0.05$)**
γ_1	14.13472514	−91.623135	+106.215141	−1.1592611550	−0.5796305775	−0.02422	7.47×10^{-11}
γ_2	21.02203964	+262.836896	−396.881347	−1.5099909961	−0.7549954980	−0.03356	3.49×10^{-10}
γ_3	25.01085758	−429.750230	+503.523871	−1.1716663211	−0.5858331605	+0.02729	2.81×10^{-10}
γ_4	30.42487613	+575.533057	−1099.753035	−1.9108425146	−0.9554212573	−0.00513	1.14×10^{-9}
γ_5	32.93506159	−700.794228	+652.362981	−0.9308909157	−0.4654454579	+0.08633	3.50×10^{-11}
γ_6	37.58617816	+1237.186036	−2075.174206	−1.6773340029	−0.8386670015	+0.02106	1.05×10^{-9}
γ_7	40.91871901	−1104.919714	+1894.851449	−1.7149222919	−0.8574611459	+0.12500	1.06×10^{-9}
γ_8	43.32707328	+1502.129769	−1349.949314	−0.8986902075	−0.4493451037	+0.10522	1.85×10^{-10}
γ_9	48.00515088	−1537.062283	+3684.290459	−2.3969688798	−1.1984844399	+0.09760	2.95×10^{-9}
γ_{10}	49.77383248	+1481.811034	−1260.292464	−0.8505082189	−0.4252541095	+0.15889	1.03×10^{-9}
γ_{11}	52.97032148	−2825.693243	+3692.728528	−1.3068398479	−0.6534199240	+0.12518	4.15×10^{-10}
γ_{12}	56.44624770	+3081.764540	−5456.621032	−1.7706158148	−0.8853079074	+0.10850	1.71×10^{-9}
γ_{13}	59.34704400	−1977.386865	+4452.836874	−2.2518794643	−1.1259397321	+0.43516	3.25×10^{-10}
γ_{14}	60.83177852	+2453.950470	−709.906506	−0.2892912938	−0.1446456469	−0.00149	6.57×10^{-11}
γ_{15}	65.11254405	−3822.839049	+8711.814953	−2.2788861476	−1.1394430738	+0.12665	3.55×10^{-9}

5.2. **Numerical tracking across $\Omega_1,\dots,\Omega_{25}$.** Table 3 summarizes predictor-corrector tracking of the first 25 imaginary nodal walls $\Gamma_1,\dots,\Gamma_{25}$ ($I(d,t) = 0$) from the true critical-line extrema τ_k ($\Xi'(\tau_k) = 0$) and real zero curves $\mathcal{R}_1,\dots,\mathcal{R}_{25}$ ($R(d,t) = 0$) from γ_k across $d \in [0, 1.65]$, passing the Euler–Stirling threshold $d = d_{\mathrm{Euler}}(t_k) > \frac{1}{2}$ in $\sigma > 1$. Along each tracked wall Γ_k, $|R|$ grows monotonically by factors between 1.81× and 6.20× up to $d = d_{\mathrm{Euler}}(t_k)$, and interlacing $t_{\Gamma_{k-1}}(d) < t_{\mathcal{R}_k}(d) < t_{\Gamma_k}(d)$ holds on all 25 tracked chambers (103,750 sampled grid points).

5.3. **Transverse integral identity, two-mechanism counterexamples, and grid scan.** Table 4 reports three sets of diagnostics for Proposition 3.2 and Theorems 4.1–4.2:

(1) *Part A (50-digit check of the boxed Transverse Integral Identity (4.1) and Proposition 3.2):* Across ten benchmark points spanning $t \in [\gamma_1, 1000]$ and $d \in [0.25, 0.50]$, Gauss–Legendre quadrature of $2u\,\widetilde{W}(u,t)$ matches $|\Xi(t - \mathrm{i}d)|^2 - |\Xi(t)|^2$ to within 8.75×10^{-51}. Simultaneously, on $\Xi_{\mathrm{hyp}}(z; d_0) = \Xi(z)(1 + (z - 17.85)^2/d_0^2)$, setting $d_0 = 0.25$ produces the boundary Laguerre violation $L_1(\Xi_{\mathrm{hyp}}; \tau_*) \approx -3.364 \times 10^{-6} < 0$ at $\tau_* \approx 17.87156653$ with no critical point in $d > 0$, whereas setting $d_0 = 1.30$ produces the interior critical point $\Xi'_{\mathrm{hyp}}(z_{\mathrm{crit}}) = 0$ at $(d_{\mathrm{crit}}, t_{\mathrm{crit}}) \approx (0.56054034, 18.66151713)$ with $L_1 > 0$ on $d = 0$.

(2) *Part B (Conditional Rolle turning point):* For $\Xi_{\mathrm{hyp}}(z) = \Xi(z)(1 + (z - t_*)^2/d_*^2)$ at $(d_*, t_*) = (0.25, 14.20)$ where $\widetilde{W}_{\mathrm{hyp}}(0, t_*)/E_0^2 = +0.549233 > 0$, $E_{\mathrm{hyp}}(d)/E_0^2$ rises from $+2.7087 \times 10^{-3}$ to an interior Rolle peak at $d_{\mathrm{peak}} \approx 0.1342212372 \in (0, 0.25)$ ($\widetilde{W}_{\mathrm{hyp}}(d_{\mathrm{peak}}, t_*) = 0$, $\partial_d \widetilde{W}_{\mathrm{hyp}}/E_0^2 = -5.834559 < 0$) before dropping to 0 via $\int_{d_{\mathrm{peak}}}^{d_*} 2u\widetilde{W}_{\mathrm{hyp}}\,\mathrm{d}u/E_0^2 = -7.182991 \times 10^{-3}$, matching $\int_0^{d_*} 2u\widetilde{W}_{\mathrm{hyp}}\,\mathrm{d}u = -|\Xi(t_*)|^2$ to within 5.22×10^{-54}.

TABLE 3. Predictor-corrector tracking and Cauchy–Riemann arc-length growth across the first 25 nodal chambers $\Omega_1, \dots, \Omega_{25}$ from $d = 0$ past $d = d_{\mathrm{Euler}}(t_k) > \frac{1}{2}$.

Wall / Chamber	**Base** τ_k	$\widehat{L}_1(\Xi;\tau_k)$	$\kappa_\Gamma(\tau_k)$	$d_{\mathrm{Euler}}(t_k)$	$t_{\Gamma_k}(1.20)$	$\frac{\lvert R(d_{\mathrm{E}})\rvert}{\lvert\Xi(\tau_k)\rvert}$	$\min \frac{\mathrm{d}\lvert R\rvert}{\mathrm{d}s}$	**Sign in** Ω_k
Γ_1 (Ω_1)	15.585709	1.5436×10^4	−0.379922	1.655324	15.310521	2.05×	$+2.25 \times 10^{-6}$	$I > 0$
Γ_2 (Ω_2)	22.097977	7.5701×10^4	−0.479403	1.442866	21.752303	2.64×	$+4.11 \times 10^{-8}$	$I < 0$
Γ_3 (Ω_3)	26.272247	2.9975×10^5	−0.406050	1.362951	25.984129	1.99×	$+2.68 \times 10^{-9}$	$I > 0$
Γ_4 (Ω_4)	31.231796	2.6560×10^5	−0.567160	1.299152	30.830599	3.65×	$+8.11 \times 10^{-11}$	$I < 0$
Γ_5 (Ω_5)	34.193310	1.0069×10^6	−0.369891	1.267179	33.936183	1.91×	$+1.03 \times 10^{-11}$	$I > 0$
Γ_6 (Ω_6)	38.498241	1.4139×10^6	−0.541509	1.232186	38.118805	2.72×	$+5.44 \times 10^{-13}$	$I < 0$
Γ_7 (Ω_7)	41.736730	1.1357×10^6	−0.512592	1.209337	41.386505	3.17×	$+4.45 \times 10^{-14}$	$I > 0$
Γ_8 (Ω_8)	44.541704	4.5977×10^6	−0.375941	1.191662	44.283576	1.88×	$+6.79 \times 10^{-15}$	$I < 0$
Γ_9 (Ω_9)	48.622533	1.5292×10^6	−0.669778	1.170877	48.165862	5.09×	$+3.04 \times 10^{-16}$	$I > 0$
Γ_{10} (Ω_{10})	50.839005	4.1899×10^6	−0.372409	1.159421	50.587875	2.20×	$+5.46 \times 10^{-17}$	$I < 0$
Γ_{12} (Ω_{12})	57.262934	8.1743×10^6	−0.577521	1.133517	56.872586	2.89×	$+5.92 \times 10^{-19}$	$I < 0$
Γ_{15} (Ω_{15})	65.758319	9.7545×10^6	−0.676205	1.105536	65.307094	4.19×	$+1.04 \times 10^{-21}$	$I > 0$
Γ_{19} (Ω_{19})	76.225438	8.8778×10^6	−0.742897	1.078266	75.748391	6.20×	$+3.40 \times 10^{-25}$	$I > 0$
Γ_{22} (Ω_{22})	83.531440	3.0440×10^7	−0.669774	1.062528	83.096458	4.19×	$+1.68 \times 10^{-27}$	$I < 0$
Γ_{25} (Ω_{25})	89.980048	1.2882×10^8	−0.321150	1.050124	89.766919	1.89×	$+1.30 \times 10^{-29}$	$I > 0$
Sampled grid check across $\Omega_1, \dots, \Omega_{25}$: $t_{\Gamma_{k-1}}(d) < t_{\mathcal{R}_k}(d) < t_{\Gamma_k}(d)$ on $(0, 1.65]$ across 103,750 interior points.								

(3) *Part C (Discrete* 60,000*-point grid scan on* $t \in [14, 1000]$*):* Evaluating 10,000 heights $t \in [14, 1000]$ across $d \in \{0.05, 0.10, 0.20, 0.35, 0.50\}$ illustrates that $\widetilde{\mathcal{E}}(d,t) = |\Xi(t - \mathrm{i}d)|^2/d^2$ exceeds $\widetilde{W}(d,t)$ on this finite grid by an average factor of 77.19× at $d = 0.05$ and 2.11× at $d = 0.35$, with $\mathcal{R}_{\mathrm{sq}}(d,t) = |\Xi(t - \mathrm{i}d)|^2/\mathcal{Q}_0(d,t) \geq 0.6544$ across the sampled points.

TABLE 4. Diagnostics of the Transverse Integral Identity (4.1), Rolle turning point (4.3), and discrete grid comparison of $\widetilde{\mathcal{E}}(d,t) = |\Xi(t-\mathrm{i}d)|^2/d^2$ versus $\widetilde{W}(d,t)$ on $t \in [14, 1000]$ (scaled by $E_0(t)^2$).

Benchmark	**Domain**	$\widehat{\Xi}(t)^2$ $(\min \widehat{\widetilde{W}})$	$d^2(\widehat{\Xi}')^2$ $(\min \widehat{\widetilde{\mathcal{E}}})$	**Integral (Mean)**	$\lvert\widehat{\Xi}\rvert^2$ $(\min \mathcal{R}_{\mathrm{sq}})$	**Residual**
Part A: Exact Transverse Integral Identity & Baseline Squares (`mp.dps = 50`)						
Zero γ_1	$(0.25, 14.1347)$	0.000000000	0.039318967	0.039485603	0.039485603	2.66×10^{-51}
Extremum τ_1	$(0.25, 15.5857)$	1.467752467	0.000000000	0.051629629	1.519382096	3.83×10^{-51}
Zero γ_2	$(0.35, 21.0220)$	0.000000000	0.158319386	0.161030413	0.161030413	3.96×10^{-51}
Extremum τ_2	$(0.35, 22.0980)$	1.123675768	0.000000000	0.148747535	1.272423303	6.85×10^{-51}
Height $t = 100$	$(0.40, 100.000)$	7.250617439	0.544807284	2.075446336	9.326063775	$< 1.00 \times 10^{-51}$
Height $t = 1000$	$(0.50, 1000.00)$	0.995594139	3.964825186	10.199646145	11.195240284	5.63×10^{-51}
Part B: Conditional Rolle Turning Point at $(d_*, t_*) = (0.25, 14.20)$ $(\widehat{E}(0), \widehat{E}(d_{\mathrm{peak}}), \int_0^{d_{\mathrm{peak}}}, \int_{d_{\mathrm{peak}}}^{d_*})$						
Perturbed Ξ_{hyp}	$d_{\mathrm{peak}} = 0.134221$	+0.002708710	+0.007182991	+0.004474281	−0.007182991	5.22×10^{-54}
True Riemann Ξ	$d \in [0, 0.25]$	+0.002708710	+0.014518426	+0.011809716	+0.028104641	$\widehat{\widetilde{W}} > 0$
Part C: Discrete Grid Scan on $t \in [14, 1000]$ $(\min \widehat{\widetilde{W}}, \min \widehat{\widetilde{\mathcal{E}}}, \textbf{Mean } \widetilde{\mathcal{E}}/\widetilde{W}, \min \mathcal{R}_{\mathrm{sq}})$						
Collar $d = 0.05$	$t \in [14, 1000]$	0.61562328	0.95174573	77.19×	0.86451727	$\mathcal{R}_{\mathrm{comp}} \geq 0.9992$
Inner $d = 0.10$	$t \in [14, 1000]$	0.61623733	0.70737264	19.76×	0.80704691	$\mathcal{R}_{\mathrm{comp}} \geq 0.9960$
Mid $d = 0.20$	$t \in [14, 1000]$	0.61869761	0.64746608	5.38×	0.71298903	$\mathcal{R}_{\mathrm{comp}} \geq 0.9629$
Outer $d = 0.35$	$t \in [14, 1000]$	0.62549700	0.63722592	2.11×	0.67256403	$\mathcal{R}_{\mathrm{comp}} \geq 0.7991$
Edge $d = 0.50$	$t \in [14, 1000]$	0.63610268	0.63926128	1.26×	0.65442574	$\mathcal{R}_{\mathrm{comp}} \geq 0.5716$

Figures 1 and 2 display the tracked nodal chambers, local curvature parabolas, conditional Rolle turning point, and parasitic nodal walls of Ξ_{hyp}.

6. Scope, limitations, and explicit obstructions

We record the logical scope of our results and the explicit analytical obstructions that prevent harmonic nodal geometry or transverse kernel filtering from yielding an unconditional proof of RH:

(1) **Circularity of critical-point-free and Laguerre-positivity hypotheses:** Propositions 2.1, 2.2(a), 3.2, and Theorems 3.1, 3.4, and 4.1 hold for *any* entire function real

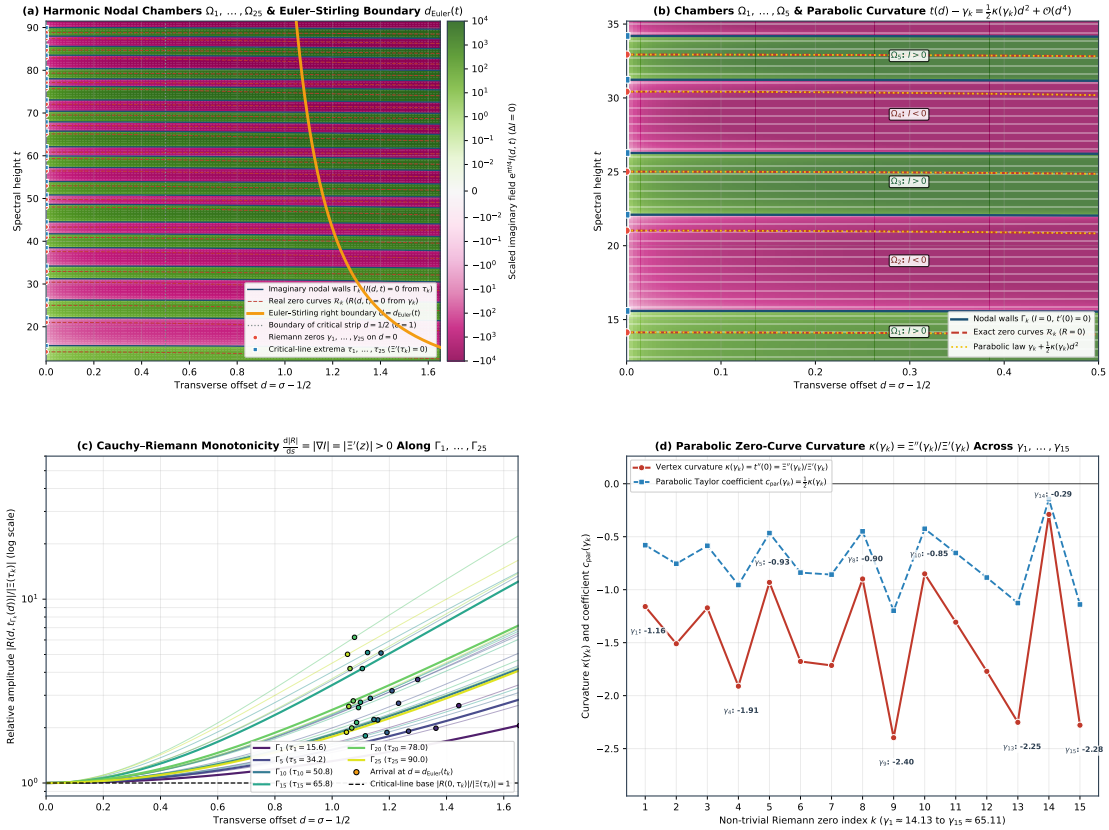


FIGURE 1. (a) Tracked harmonic nodal chambers $\Omega_1, \ldots, \Omega_{25}$ from $d = 0$ across $d = \frac{1}{2}$ ($\sigma = 1$) to the Euler–Stirling boundary $d = d_{\text{Euler}}(t) > \frac{1}{2}$. (b) Close-up of $\Omega_1, \ldots, \Omega_5$ showing orthogonal departure of Γ_k from τ_k and agreement between $\mathcal{R}_k$ and $t(d) = \gamma_k + \frac{1}{2}\kappa(\gamma_k)d^2$. (c) Arc-length growth $|R(d, t_{\Gamma_k}(d))|/|\Xi(\tau_k)|$ along $\Gamma_1, \ldots, \Gamma_{25}$. (d) Zero-curve curvature $\kappa(\gamma_k) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$ and $c_{\text{par}}(\gamma_k) = \frac{1}{2}\kappa(\gamma_k)$ across $\gamma_1, \ldots, \gamma_{15}$.

on $\mathbb{R}$. To conclude globally that $|R|$ grows along every wall Γ_k, one must assume both $L_1(\Xi; \tau_k) > 0$ on $d = 0$ and $\Xi'(t - \mathrm{i}d) \neq 0$ on $d > 0$; by Lagarias [10], the latter hypothesis is already equivalent to RH.

(2) **The Davenport–Heilbronn obstruction:** The completed Davenport–Heilbronn function $\Xi_{\text{DH}}(z)$ [5, 24] is entire of order 1, real on $\mathbb{R}$, symmetric under $s \leftrightarrow 1 - s$, harmonic ($\Delta R_{\text{DH}} = \Delta I_{\text{DH}} = 0$), and satisfies the arc-length equation (3.1) and the boxed Transverse Integral Identity (4.1). Nevertheless, lacking an Euler product, Ξ_{DH} has infinitely many zeros in $d > 0$ (including in $\sigma > 1$), realized geometrically through boundary extrema with $L_1(\Xi_{\text{DH}}; \tau) < 0$ and interior critical points $\Xi'_{\text{DH}}(z_{\text{crit}}) = 0$ per Proposition 3.2.

(3) **Sign oscillations of $I_{\text{Wig}}(y, t)$ and Lehmer/Skewes phenomena:** Although Theorem 4.2(a) maps $\text{sinc}_{\text{hyp}}(2uy)$ into the non-negative square $\text{sinc}^2_{\text{hyp}}(dy) \geq 1$ and damps large shifts by $\mathcal{F}_{\text{int}}(2dy) = \dfrac{\tanh(dy)}{dy} \in (0, 1)$, the Wigner correlation kernel $I_{\text{Wig}}(y, t)$ changes sign rapidly with y (exhibiting 5, 6, and 7 sign changes already at $\gamma_3, \gamma_4, \gamma_5$ [12]). Hence positivity of $\int_0^\infty y^2 \,\text{sinc}^2_{\text{hyp}}(dy) I_{\text{Wig}}(y, t)\,\mathrm{d}y$ cannot be deduced from integrand positivity, and at extreme Lehmer pairs or Skewes-scale phase alignments where $|\Xi'(\gamma_k)|$ is anomalously small, $|\kappa(\gamma_k)|$ becomes arbitrarily large while the baseline square $\mathcal{Q}_0(d, \gamma_k) = d^2(\Xi'(\gamma_k))^2$ becomes arbitrarily small.

(4) **Restriction of $d_{\text{Euler}}(t)$ to $\sigma > 1$ and Voronin universality in $\frac{1}{2} < \sigma < 1$:** The deductive boundary $d_{\text{Euler}}(t) > \frac{1}{2}$ in Proposition 2.2(b) lies strictly inside the Euler-product half-plane $\sigma = \frac{1}{2} + d > 1$, approaching $\sigma = 1^+$ only at rate $(\frac{1}{2}\ln t)^{-1}$. Inside

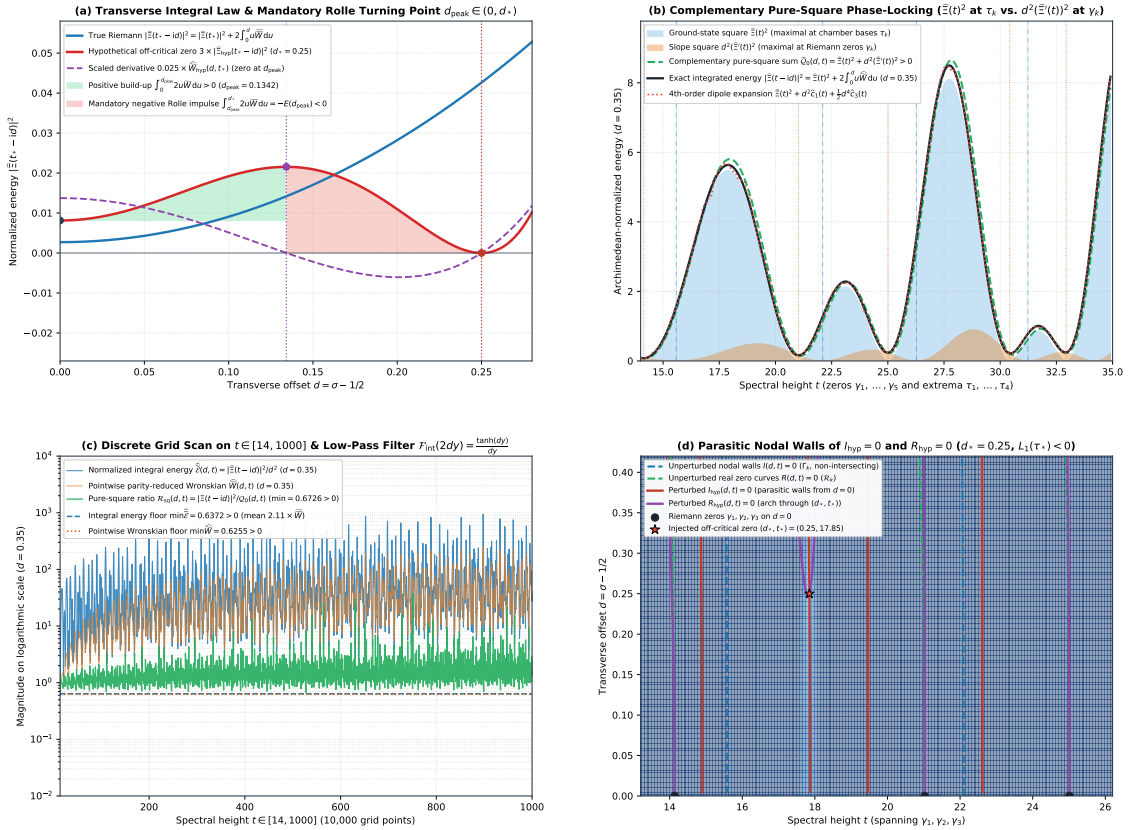


FIGURE 2. (a) Comparison of $|\widehat{\Xi}(t_* - \mathrm{i}d)|^2$ against the off-critical perturbation $|\widehat{\Xi}_{\text{hyp}}(t_* - \mathrm{i}d)|^2$ ($d_* = 0.25, t_* = 14.20$), illustrating the Rolle turning point $d_{\text{peak}} \in (0, d_*)$. (b) Quadrature interleaving of $\widehat{\Xi}(t)^2$ (maximal at τ_k) and $d^2(\widehat{\Xi}'(t))^2$ (maximal at γ_k) on $t \in [14, 35]$. (c) Discrete grid scan on $t \in [14, 1000]$ at $d = 0.35$. (d) Parasitic nodal walls of $I_{\text{hyp}} = 0$ and arch of $R_{\text{hyp}} = 0$ for $\Xi_{\text{hyp}}(z; 0.25)$, where $L_1(\Xi_{\text{hyp}}; \tau_*) < 0$ on $d = 0$.

the open right half of the critical strip $0 < d < \frac{1}{2}$ ($\frac{1}{2} < \sigma < 1$), the Dirichlet series for $-\zeta'(s)/\zeta(s)$ diverges and Voronin's universality theorem [26] implies that $\zeta'(s)/\zeta(s)$ approximates arbitrary complex values on vertical translates, precluding any triangle-inequality majorant of the form (2.7) for $d < \frac{1}{2}$.

APPENDIX A. QUARTIC NODAL-CURVE JET AND EULER–STIRLING ASYMPTOTICS

Lemma A.1 (Quartic nodal-curve coefficients and asymptotic expansion of $d_{\text{Euler}}(t)$)**.** (a) *At any simple real zero γ_k ($f(\gamma_k) = 0$, $f'(\gamma_k) \neq 0$) of an entire function f real on $\mathbb{R}$, the fourth-order coefficient $c_{\text{quart}}(\gamma_k) = \frac{1}{24}t^{(4)}_{\mathcal{R}_k}(0)$ in (3.4) is given by (3.5). At any non-degenerate real critical point τ_k ($f'(\tau_k) = 0$, $f''(\tau_k) \neq 0$), the imaginary nodal curve Γ_k satisfies $t_{\Gamma_k}(d) - \tau_k = c_{\text{par},\Gamma}(\tau_k)d^2 + c_{\text{quart},\Gamma}(\tau_k)d^4 + \mathcal{O}(d^6)$ with $c_{\text{par},\Gamma}(\tau_k) = \frac{f'''(\tau_k)}{6f''(\tau_k)}$ and*

$$c_{\text{quart},\Gamma}(\tau_k) = \frac{20\, f^{(4)}(\tau_k)\, c_{\text{par},\Gamma}(\tau_k) - 60\, f'''(\tau_k)\, c_{\text{par},\Gamma}(\tau_k)^2 - f^{(5)}(\tau_k)}{120\, f''(\tau_k)}. \tag{A.1}$$

(b) *As $t \to \infty$, the root $d_{\text{Euler}}(t) > \frac{1}{2}$ solving $\mathcal{A}(d,t) = M_{\text{Euler}}(d)$ in Proposition 2.2(b) satisfies*

$$d_{\text{Euler}}(t) = \frac{1}{2} + \frac{1}{\frac{1}{2}\ln\frac{t}{2\pi} + \gamma_{\text{E}} + \mathcal{O}((\ln t)^{-1})} \longrightarrow \frac{1}{2}^{+}, \tag{A.2}$$

where $\gamma_{\mathrm{E}} = 0.5772156649\ldots$ *is the Euler–Mascheroni constant.*

Proof. (a) Truncating (2.5) at d^4 gives $R_f(d,t) = f(t) - \frac{1}{2}f''(t)d^2 + \frac{1}{24}f^{(4)}(t)d^4 + \mathcal{O}(d^6) = 0$. Substituting $t = \gamma_k + c_2 d^2 + c_4 d^4 + \mathcal{O}(d^6)$ around $f(\gamma_k) = 0$ yields

$$\Big[f'(\gamma_k)(c_2 d^2 + c_4 d^4) + \frac{1}{2}f''(\gamma_k)c_2^2 d^4\Big] - \frac{1}{2}\Big[f''(\gamma_k) + f'''(\gamma_k)c_2 d^2\Big]d^2 + \frac{1}{24}f^{(4)}(\gamma_k)d^4 + \mathcal{O}(d^6) = 0.$$

Matching d^2 gives $c_2 = c_{\mathrm{par}}(\gamma_k) = \frac{f''(\gamma_k)}{2f'(\gamma_k)}$, and matching d^4 gives $f'(\gamma_k)c_4 + \frac{1}{2}f''(\gamma_k)c_2^2 - \frac{1}{2}f'''(\gamma_k)c_2 + \frac{1}{24}f^{(4)}(\gamma_k) = 0$, proving (3.5). Substituting $t = \tau_k + c_{2,\Gamma}d^2 + c_{4,\Gamma}d^4 + \mathcal{O}(d^6)$ into $\tilde{I}_f(d,t) = -f'(t) + \frac{1}{6}f'''(t)d^2 - \frac{1}{120}f^{(5)}(t)d^4 + \mathcal{O}(d^6) = 0$ similarly yields (A.1).

(b) By Stirling's expansion $\psi(z) = \ln z - \frac{1}{2z} + \mathcal{O}(|z|^{-2})$ as $\mathrm{Im}(z) \to \infty$, $\mathcal{A}(d,t) = \frac{1}{2}\ln\frac{t}{2\pi} + \mathcal{O}(t^{-2})$ uniformly on $d \in [\frac{1}{2}, 2]$. Writing $\sigma = \frac{1}{2} + d = 1 + \varepsilon$ ($\varepsilon > 0$), the Laurent expansion $-\frac{\zeta'(1+\varepsilon)}{\zeta(1+\varepsilon)} = \frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \mathcal{O}(\varepsilon)$ [24, Section 2.1] gives $\frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \mathcal{O}(\varepsilon) = \frac{1}{2}\ln\frac{t}{2\pi} + \mathcal{O}(t^{-2})$, proving (A.2). □

Use of AI and Computational Tools

Gemini (Google DeepMind) was used interactively to assist with symbolic manipulations of the fourth-order implicit nodal-curve jets, literature cross-checks, and generation of the `mpmath`, `mpmath.iv`, and GPU verification scripts in `scripts/verify_paper3_harmonic_nodal_chambers.py`. All analytical proofs, interval bounds, and bibliographic references were verified independently.

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